Synchronous FIFO Report

A. Encountered Bugs and Timing Violations

1. Monitor & Interface Timing Bugs

Bug 1.1: Zero-Cycle Read Sampling (Stale Data Capture)

```
[80] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x5 || full : 0 || empty : 0
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 80: reporter [MISCMP] exp_item@680 vs. m_item@676
UVM_INFO scoreboard.sv(56) @ 80: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH : DOUT - 0x5 (EXP) vs 0x0 (ACT) || FULL - 0 vs 0 || EMPTY
- 0 vs 0
[90] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x5 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(53) @ 90: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH : DOUT - 0x5
[100] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x3 || full : 0 || empty : 0
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 100: reporter [MISCMP] exp_item@702 vs. m_item@698
UVM_INFO scoreboard.sv(56) @ 100: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH : DOUT - 0x3 (EXP) vs 0x5 (ACT) || FULL - 0 vs 0 || EMPTY
 - 0 vs 0
```

- **Symptom:** The Scoreboard logged actual data as 0x0 during the first read, followed by subsequent reads being off-by-one compared to the Predictor.
- **Root Cause:** The synchronous FIFO has a 1-clock-cycle read latency. The Output Monitor evaluated rd_en and sampled dout on the exact same clock edge, capturing the stale data left on the bus before the RTL had time to fetch the new data.
- **Solution:** Pipelined the monitor. The monitor was restructured to evaluate read requests in the *current* cycle but wait until the *next* clock cycle to sample dout and broadcast to the Scoreboard.

Bug 1.2: Blocking Waits Stalling the Pipeline

```
[30] [INPUT MONITOR] write : 1 || din : 0x7
[40] [INPUT MONITOR] write : 1 || din : 0x7
[50] [INPUT MONITOR] write : 1 || din : 0x4
[60] [INPUT MONITOR] write : 1 || din : 0x4
[70] [INPUT MONITOR] write : 1 || din : 0x1
[80] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x7 || full : 0 || empty : 0
[90] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x7 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(53) @ 90: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH : DOUT - 0x7
[100] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x4 || full : 0 || empty : 0

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```

- **Symptom:** The Output Monitor successfully captured the first read but completely missed back-to-back read transactions, causing the Scoreboard to only print once.
- **Root Cause:** The monitor used multiple blocking clock delays (@(posedge clk)) inside a single forever loop iteration. This consumed two clock cycles to process a single transaction, causing the monitor to be "blind" to the bus during consecutive reads.

- **Solution:** Decoupled the "read check" from the "data fetch" using a non-blocking read_pending state flag, allowing the monitor's forever loop to evaluate the bus on every single clock edge without stalling.

2. Reference Model (Predictor) State Divergence

Bug 2.1: Concurrent Underflow Misprediction

```
UVM_INFO scoreboard.sv(49) @ 170: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0xbb
[180] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x5 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 180: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0xbb
[190] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x4 || full : 0 || empty : 0
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 190: reporter [MISCMP] exp_item@760 vs. m_item@774
UVM_INFO scoreboard.sv(52) @ 190: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x2: DOUT - 0x5 (EXP) vs 0xbb (ACT) || FULL - 0 vs 0 |
| EMPTY - 0 vs 0
[200] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x7 || full : 0 || empty : 0
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 200: reporter [MISCMP] exp_item@770 vs. m_item@786
UVM_INFO scoreboard.sv(52) @ 200: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x2: DOUT - 0x4 (EXP) vs 0x5 (ACT) || FULL - 0 vs 0 ||
 EMPTY - 0 vs 0
[210] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x3 || full : 0 || empty : 1
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 210: reporter [MISCMP] exp_item@782 vs. m_item@798
UVM_INFO scoreboard.sv(52) @ 210: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x2: DOUT - 0x7 (EXP) vs 0x4 (ACT) || FULL - 0 vs 0 ||
 EMPTY - 0 vs 0
[220] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> EMPTY
UVM_INFO scoreboard.sv(60) @ 220: uvm_test_top.env.scb [Scoreboard] FIFO IS EMPTY
UVM_INFO scoreboard.sv(60) @ 230: uvm_test_top.env.scb [Scoreboard] FIFO IS EMPTY

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```

- **Symptom:** A mismatch occurred during a simultaneous Read/Write operation when the FIFO was perfectly empty. The Scoreboard expected 0x0 but the RTL output the previous transaction's data.
- **Root Cause:** The RTL correctly evaluated !empty (false) and ignored the read request while executing the write. However, the Predictor processed the write *first*, meaning it popped the newly written data instead of ignoring the read. This caused the Predictor's internal queue to permanently shift one transaction ahead of the RTL.
- **Solution:** Enforced strict operation ordering. The Predictor was updated to evaluate the validity of read/write requests against the *current* queue state before executing either operation, perfectly mimicking the RTL's synchronous behavior.

Bug 2.2: Mid-Cycle Flag Evaluation (Time-of-Evaluation Trap)

```
[PREDICTOR] Pop Expected -> dout : 0x8 || full : 0 || empty : 1
UVM_INFO scoreboard.sv(49) @ 330: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x56
[340] [INPUT MONITOR] write : 1 || read : 1 || din : 0x67
[PREDICTOR] Pop Expected -> EMPTY
[PREDICTOR] Push Expected At 0x1 -> din : 0x67
UVM_INFO scoreboard.sv(60) @ 340: uvm_test_top.env.scb [Scoreboard] FIFO IS EMPTY
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 350: reporter [MISCMP] exp_item@856 vs. m_item@876
UVM_INFO scoreboard.sv(52) @ 350: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x1: DOUT - 0x8 (EXP) vs 0x8 (ACT) || FULL - 0 vs 0 || EMPTY - 1 vs 0
[360] [INPUT MONITOR] write : 1 || read : 1 || din : 0x9d
[PREDICTOR] Pop Expected -> dout : 0x67 || full : 0 || empty : 1
[PREDICTOR] Push Expected At 0x1 -> din : 0x9d
[370] [INPUT MONITOR] write : 1 || read : 1 || din : 0x9e
[PREDICTOR] Pop Expected -> dout : 0x9d || full : 0 || empty : 1
[PREDICTOR] Push Expected At 0x1 -> din : 0x9e
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 370: reporter [MISCMP] exp_item@884 vs. m_item@894
UVM_INFO scoreboard.sv(52) @ 370: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x2: DOUT - 0x67 (EXP) vs 0x67 (ACT) || FULL - 0 vs 0 || EMPTY - 1 vs 0
[380] [INPUT MONITOR] write : 1 || read : 1 || din : 0xc1
[PREDICTOR] Pop Expected -> dout : 0x9e || full : 0 || empty : 1
[PREDICTOR] Push Expected At 0x1 -> din : 0xc1
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 380: reporter [MISCMP] exp_item@890 vs. m_item@906
UVM_INFO scoreboard.sv(52) @ 380: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x2: DOUT - 0x9d (EXP) vs 0x9d (ACT) || FULL - 0 vs 0 || EMPTY - 1 vs 0
[390] [INPUT MONITOR] write : 1 || din : 0xc3
[PREDICTOR] Push Expected At 0x2 -> din : 0xc3
UVM_INFO /home/thanhtai/Libraries/uvm-verilator/src/base/uvm_comparer.svh(527) @ 390: reporter [MISCMP] exp_item@902 vs. m_item@916
UVM_INFO scoreboard.sv(52) @ 390: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MISMATCH AT 0x1: DOUT - 0x9e (EXP) vs 0x9e (ACT) || FULL - 0 vs 0 || EMPTY - 1 vs 0
[400] [INPUT MONITOR] write : 1 || din : 0x2d
[PREDICTOR] Push Expected At 0x3 -> din : 0x2d
```

- **Symptom:** During valid concurrent read/writes, the Predictor broadcasted empty = 1, while the RTL broadcasted empty = 0, causing a scoreboard mismatch despite the data payload matching perfectly.
- **Root Cause:** The Predictor recalculated the empty flag sequentially *between* the pop and the push. It temporarily saw an empty queue and broadcasted the state. The RTL evaluated both operations simultaneously, recognizing the count remained at 1.
- **Solution:** Restructured the Predictor to execute both the pop and the push, and only calculate the expected full/empty flags based on the *final* size of the internal queue at the end of the function.

3. Scoreboard Synchronization Traps

Bug 3.1: Valid Data Dropped on Empty Transition

```
[PREDICTOR] Push Expected At 0x6 -> din : 0x2
[80] [INPUT MONITOR] write : 1 || din : 0x3
[PREDICTOR] Push Expected At 0x7 -> din : 0x3
[90] [INPUT MONITOR] write : 1 || din : 0x4
[PREDICTOR] Push Expected At 0x8 -> din : 0x4
[100] [INPUT MONITOR] write : 1 || din : 0x0
[PREDICTOR] Push Expected -> FULL
[110] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x0 || full : 0 || empty : 0
[120] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x7 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 120: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x0
[130] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x5 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 130: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x7
[140] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x6 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 140: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x5
[150] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x1 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 150: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x6
[160] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x2 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 160: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x1
[170] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x3 || full : 0 || empty : 0
UVM_INFO scoreboard.sv(49) @ 170: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x2
[180] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x4 || full : 0 || empty : 1
UVM_INFO scoreboard.sv(49) @ 180: uvm_test_top.env.scb [Scoreboard] [SCB] DATA MATCH AT 0x2 : DOUT - 0x3
[190] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> EMPTY
UVM_INFO scoreboard.sv(60) @ 190: uvm_test_top.env.scb [Scoreboard] FIFO IS EMPTY
UVM_INFO scoreboard.sv(60) @ 200: uvm_test_top.env.scb [Scoreboard] FIFO IS EMPTY
```

- **Symptom:** The 8th read transaction in an 8-deep FIFO vanished from the Scoreboard logs, replaced by a "FIFO IS EMPTY" bypass message.
- **Root Cause:** Custom bypass logic inside the Scoreboard dropped transactions if item.empty == 1. When a valid read causes the FIFO to empty, dout is still perfectly valid.
- **Solution:** Removed the bypass. Configured the Scoreboard to blindly trust the queue size. If the Predictor queued an item, the Scoreboard must pop and compare it, regardless of the flags attached to the actual item.

4. SystemVerilog Assertions & Sequence Driving

Bug 4.1: Overlapping Implication on Pointer Stability (False Failure)

```
[80] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x6 || full : 0 || empty : 0
[90] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x6 || full : 0 || empty : 0
[100] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x0 || full : 0 || empty : 0
[110] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x0 || full : 0 || empty : 0
[120] [INPUT MONITOR] read : 1
[PREDICTOR] Pop Expected -> dout : 0x7 || full : 0 || empty : 1
[130] %Error: assertion.sv:53: Assertion failed in TOP.top.dut_wrapper.DUT.a0.a_read_pointer_empty: 'assert' failed.
%Error: assertion.sv:53: Verilog $stop
Aborting...
make: *** [Makefile:56: run] Error 1
```

- **Symptom:** A valid SVA check (rd_en && empty) |-> $stable(rd_pt) threw a failure on the exact cycle the FIFO transitioned to an empty state.
- **Root Cause:** SVA evaluates in the Preponed region. Using the overlapping implication (|->) checked the pointer stability in the exact same cycle the read finished, flagging the legitimate increment from the previous cycle as an error.
- **Solution:** Swapped to the non-overlapping implication (|=>), properly ensuring that the pointer remains stable in the cycle *after* the FIFO is detected as empty.

Bug 4.2: Frozen Bus States (Sequence Underflow)

- **Symptom:** The assertion failure above was triggered because the driver attempted to read an empty FIFO long after the sequence had finished executing.
- **Root Cause:** The UVM Driver utilized the blocking get_next_item() task. When the sequence ended, the driver froze indefinitely, leaving rd_en permanently pulled high at 1.
- **Solution:** Replaced the blocking call with the non-blocking try_next_item() task. This allowed the driver to actively detect an empty sequence queue and drive the physical interface back to a safe, idle state (wr_en = 0, rd_en = 0).

B. Functional Coverage

1. **Number of tests :** 50
2. **Percentage of coverage :** 100 %

```
============================================================
               CUMULATIVE COVERAGE SUMMARY
============================================================
 Total Tests Executed : 50
 Passing Tests        : 50/50
------------------------------------------------------------
 [HIT] Empty                             : 152 hits
 [HIT] Near Empty                        : 204 hits
 [HIT] Mid Range                         : 661 hits
 [HIT] Near Full                         : 85 hits
 [HIT] Full                              : 85 hits
 [HIT] Write when Full (Overflow)         : 43 hits
 [HIT] Read when Empty (Underflow)       : 71 hits
 [HIT] Simultaneous RW (Empty)           : 30 hits
 [HIT] Simultaneous RW (Mid)             : 157 hits
 [HIT] Simultaneous RW (Full)            : 12 hits
 [HIT] Reset at Full Capacity            : 10 hits
 [HIT] Reset at Empty Capacity           : 10 hits
============================================================
 FINAL FUNCTIONAL COVERAGE: 100.0%
============================================================
```